

RFI mitigation algorithm based on adaptive filter

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ABSTRACT

Aiming at the problem of radio observation data quality degradation or invalid data caused by Radio Frequency Interference (RFI), we have studied the adaptive RFI algorithm and proposed an interference mitigation method based on the adaptive filter. An adaptive interference suppression model has been established to effectively counter complex and intense interference. This model includes a normalized factor, derived from the filter's correlation function, which improves the performance of the adaptive filtering algorithm in reducing RFI. The adaptive filter algorithm flexibly adjusts the filter gain automatically, based on the correlation between the RFI reference signal and the observation signal, thereby effectively eliminating the RFI mixed with astronomical signals. Simulation results show that the algorithm can effectively eliminate radio frequency interference, improve the signal-to-noise ratio of astronomical signals and signal recognition rate. It also prevents the loss of signal bandwidth, ensuring the provision of reliable data for subsequent post-processing analysis.

Keywords: RFI, Adaptive Filter, Reference Antenna

1. INTRODUCTION

Radio telescopes, with their large apertures and high sensitivity, capture faint astronomical signals but are inevitably affected by radio frequency interference (RFI). RFI primarily originates from human communication and daily activities [1,2], significantly impacting the sensitivity and data quality of astronomical equipment. It has the potential to degrade data quality, cause receiver saturation, and consequently, compromise the precision and reliability of scientific research outcomes. To achieve their expected scientific goals, high-sensitivity, large-aperture radio observation equipment requires effective RFI protection and suppression measures. Therefore, researching RFI identification and suppression technologies is essential, and how to quickly and accurately identify and suppress RFI has become an urgent problem to solve in radio observation.

Signals received by radio telescopes worldwide are affected by RFI to varying degrees, and the increasing demand for the radio spectrum from commercial, defense, and other interested institutions further limits the resources available for astronomical observations. The reliance on radio regulatory agencies alone is insufficient to ensure interference-free observations. Consequently, radio astronomers and engineers must research and develop effective methods to suppress interference in the presence of astronomical signals [3].

Observation data directly impacts scientific achievements and the credibility of conclusions. RFI suppression and mitigation technologies are continually advancing to address these scientific demands. Prominent astronomical signal processing systems worldwide, such as Australia's FPGA [4] and the FPGA+GPU systems [5] developed by CASPER in the United States, have integrated methods for RFI detection and suppression. Reference [6] details an adaptive RFI cancellation method that uses a reference antenna to recover original astronomical signals from interference-affected data. References [3] and [7] report modifications to the adaptive filter algorithm, which were later tested using the Parkes radio telescope [8].

However, the increasing complexity of the RFI environment presents substantial challenges to RFI suppression technology. Existing algorithms struggle to meet the real-time processing requirements for ultra-wideband, high-speed signals. Adaptive filters hold practical value and application prospects in the radio astronomy field. Research on relevant algorithms

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and technologies has been actively conducted both domestically and internationally. Australia is at the forefront of these efforts, with ATNF promoting the development of GPU cluster adaptive filters and related research, which is poised to be implemented in the ultra-wideband receiving system of the Parkes radio telescope [9,10]. This study analyzes adaptive filters and proposes a novel adaptive RFI suppression algorithm.

2. ADAPTIVE FILTER THEORY

The adaptive interference suppression system dynamically adjusts the filter tap coefficients in real time based on signals from the RFI reference antenna, effectively reducing the RFI that is mixed into astronomical signals. The adaptive filter [11] is the core component of the RFI suppression system. By modifying the amplitude and phase of the reference antenna signal and subtracting the corresponding interference components from the astronomical data, the system can obtain signals with reduced RFI contamination. The main advantage of adaptive filters is their ability to suppress RFI while causing minimal disturbance to the astronomical data, thus enhancing the quality of the observation signals and minimizing data loss. This characteristic makes adaptive filters the optimal choice for interference suppression during radio observations. The adaptive interference suppression system can provide reliable observation data for further processing, such as pulsar data de-dispersion, folding, and timing. The model for the adaptive RFI suppression algorithm is illustrated in Figure 1.

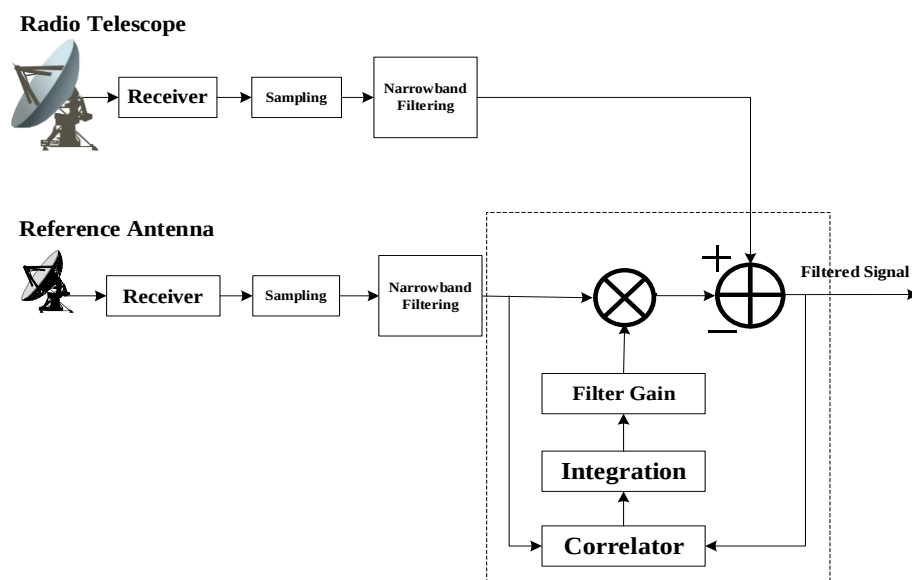


Figure 1. Adaptive RFI Elimination Algorithm Model.

Both the reference antenna and observing telescope can receive the signal from the same interference source, so there is a high correlation between the RFI components in the reference and observation signals. This correlation can be leveraged to distinguish between the two signal sources. As radio telescopes track observed sources, the intensity of RFI may fluctuate over time, leading to significant variations in RFI intensity and phase between the two signals. We employed cross-correlation methods to analyze these signals and found that their characteristics are related to the interference frequency. When the interference source is in the same direction as the observed source, the radio telescope will experience the highest RFI intensity, which is at its lowest when the interference source is in the opposite direction. RFI suppression is necessary to separate the interference from the target signal. Adaptive filters use gain functions to adjust the strength or phase of RFI reference signals to match the RFI component in the astronomical signals, and then suppress RFI by subtracting the two. The adaptive filtering algorithm can effectively suppress different frequency components. Adaptive filters only affect correlated signals and do not alter the characteristics of astronomical signals, such as those from pulsars, while maintaining the signal bandwidth unchanged.

Adaptive RFI suppression algorithms should be executed on narrowband channels of astronomical signals to suppress RFI better. The baseband astronomical signal is divided into multiple channels using narrowband filters or fast channel splitting techniques, allowing for independent RFI suppression in each channel. This approach not only improves the efficiency of interference suppression but also enables flexible channel selection for RFI mitigation based on the quality of the signal

within each channel. Currently, common channelization methods include fast Fourier transform (FFT) or polyphase filter banks (PFB) for fast channel partitioning.

The adaptive filter performs RFI suppression based on the correlation between the input signal and the reference signal, and it needs to ensure that the input signal achieves a synchronized state beforehand. If there is a delay between the astronomical signal and the RFI reference signal, the filter's anti-interference ability will significantly weaken. Therefore, signal synchronization is necessary before introducing the baseband signal into the adaptive filter.

The time delay between astronomical signals and RFI reference signals is inevitable. The different air propagation paths of interference signals from the RFI source to the main and reference antenna resulted in a delay in the received signal. Therefore, when analyzing RF front-end equipment, it is necessary to consider the temporal discrepancy between the working state of the RF front-end equipment and the measured object. Additionally, various electronic devices, analog-to-digital conversion systems, and different-length optical cables used between the digital backend systems and the radio telescope and reference antennas can introduce delays in the input signals to adaptive filters.

Building on this theoretical foundation, we delve into the causes of RFI and present techniques for its suppression. Utilizing signal processing methods such as matched filters and correlators, we detect signal delays to enhance the performance of the RFI processing system. Informed by experimental data on signal delays, we can achieve real-time synchronization of baseband signals through the design of a digital signal preprocessing module. Finally, we will get the achievement of RFI elimination based on adaptive filtering.

3. ALGORITHM DESIGN OPTIMIZATION

The principle of RFI suppression using adaptive filters, as introduced earlier, is based on the correlation between the input signal and the reference signal. Assuming there is a pulse signal from pulsars mixed up with RFI.

The RFI reference antenna signal can be represented as:

$$V_r(t) = V_{r_rx}(t) + V_{r_rfi}(t) + V_{r_ast}(t) \quad (1)$$

Where $V_{r_rx}(t)$ is the receiver noise of the reference antenna system. The $V_{r_rfi}(t)$ and $V_{r_ast}(t)$ are the RFI and pulsar signal received by the reference antenna, respectively.

The observation antenna signal can be represented as:

$$V_a(t) = V_{a_rx}(t) + V_{a_rfi}(t) + V_{a_ast}(t) \quad (2)$$

Where $V_{a_rx}(t)$ is the receiver noise of the observation system. The $V_{a_rfi}(t)$ and $V_{a_ast}(t)$ are the RFI signal and pulsar signal received by the observation antenna, respectively.

Perform cross-correlation operation between the output of the adaptive filter and the reference antenna signal:

$$C(g) = \langle (V_a - gV_r) \cdot (V_r) \rangle \quad (3)$$

Where g is the gain function of the adaptive filter, and the filter gain adjusts the correlation to zero through a feedback system. Therefore, when $C(g)=0$.

$$g = \frac{\langle V_{a_rfi} V_{r_rfi} \rangle + \langle V_{a_ast} V_{r_ast} \rangle}{V_{r_rx}^2 + V_{r_rfi}^2 + V_{r_ast}^2} \quad (4)$$

Given the small aperture and low sensitivity of the reference antenna, it will receive an exceedingly weak pulsar signal. Consequently, we can approximate that $V_{r_ast}(t)=0$, allowing us to simplify the expression as follows:

$$g = \frac{\langle V_{a_rfi} V_{r_rfi} \rangle}{V_{r_rx}^2 + V_{r_rfi}^2} \quad (5)$$

assuming $V_{a_rfi} = \xi V_{r_rfi}$, we can obtain:

$$g = \frac{\xi}{1 + 1/INR} \quad (6)$$

Where INR stands for the signal-to-noise ratio in the reference signal, $INR = V_{r_rfi}^2 / V_{r_rx}^2$.

The adaptive filter iteratively generates the optimal filter gain from cross-correlation operations, which can be expressed as:

$$g_n = g_{n-1} + \varepsilon \frac{1}{\tau} \int_{t-\tau/2}^{t+\tau/2} (V_a(t) - g_{n-1} V_r(t)) V_r^*(t) dt \quad (7)$$

Where τ is the filter gain integration time. The adaptive filter continuously corrects and controls the phase and gain of the reference signal until the RFI component matches the RFI in the astronomical signal.

The output function of the adaptive filter is:

$$\begin{aligned} V_{filt} &= V_a - g V_r \\ &= V_{a_rx} + (V_{a_rfi} - g V_{r_rfi}) + (V_{a_ast} - g V_{r_ast}) - g V_{r_rx} \end{aligned} \quad (8)$$

Data block processing techniques can be used to optimize interference suppression algorithms' performance and execution speed. This method can significantly reduce the algorithm's computational requirements, thereby improving RFI elimination efficiency. By leveraging data block processing, we can fully utilize parallel computing resources to achieve real-time multi-threaded interference suppression. Building on this, we propose an improved algorithm to accelerate computational efficiency and ensure filtering accuracy. Firstly, we split the raw data into multiple independent data blocks. Then, each data block is responded to and updated by the filter, allowing for parallel processing. The length of these data blocks is correlated with the filtering effects achieved under identical conditions. The size of a data block is directly proportional to its parallel processing performance. However, as the length of the data block grows, the convergence time of the adaptive filter extends. Therefore, the data block size needs to be adjusted according to specific requirements.

The fixed constant ε in the iterative formula of the adaptive filter tap coefficient represents each iteration step size. It can affect the filter convergence time and the RFI suppression algorithm performance. If ε is too small, the adaptive filter will take longer to reach the stable state; If ε is too large, it will cause the output data to diverge and result in incorrect results. Generally, ε is a complex number less than one and approximately equal to $1/RFI$. The dependence of ε on the input signal amplitude can be solved by introducing the normalization factor of the filter cross-correlation function.

The average power of one data block in both astronomical signals and reference antenna signals can be expressed as:

$$P_{ref}(n) = \frac{1}{k} \sum_{i=0}^{k-1} V_{ref}(i + n \times k) \times V_{ref}^*(i + n \times k) \quad (9)$$

$$P_{ast}(n) = \frac{1}{k} \sum_{i=0}^{k-1} V_{ast}(i + n \times k) \times V_{ast}^*(i + n \times k) \quad (10)$$

Where $P_{ref}(n)$ and $P_{ast}(n)$ are each data block average power in the reference antenna signal and astronomical signal, n is the data block index, i is the sampling index, and k is the data block length. The data block size can be modified according to actual application requirements.

The standardized parameter $f(n)$ is as follows:

$$f(n) = P_{ast}(n) + P_{ref}(n) \quad (11)$$

$$f(n) = \begin{cases} f(n), & (n = 0) \\ (1-u) \times f(n-1) + u \times f(n), & (n > 0) \end{cases} \quad (12)$$

Where u is the update factor and can be taken according to actual needs.

Using standardized parameters, the cross-correlation function between the adaptive filter output and the reference signal can be recalculated:

$$C(i) = \frac{(V_{ast}(i) - g(n) \times V_{ref}(i)) \times V_{ref}^*(i)}{f(n)} \quad (13)$$

The average value of cross-correlation is:

$$C_{av}(n) = \frac{1}{k} \sum_{i=0}^{k-1} C(i + n \times k) \quad (14)$$

The new adaptive filter gain function is:

$$g(n) = g(n-1) + \varepsilon \times C_{av}(n) \quad (15)$$

Optimized algorithms mitigate the dependency between the convergence factor ε and the amplitude of the input signal, thereby significantly enhancing the performance of the adaptive anti-interference algorithm. By employing multithreading, a higher acceleration ratio can be realized, enabling the implementation of the RFI elimination algorithm. The proposed algorithm is capable of tackling the substantial computational challenges that arise during the real-time interference cancellation process.

4. EXPERIMENTAL RESULTS AND ANALYSIS

A series of simulation experiments were conducted to verify the performance of the adaptive filtering algorithm in RFI elimination. The simulations generated two sets of signals: astronomical and RFI reference signals. The astronomical signal consists of three components: white noise, an RFI signal, and a pulsar signal. The RFI reference signal primarily includes white noise and interference signals and does not include the pulsar signal component. Due to the small aperture and low sensitivity of the reference antenna, along with its main lobe directed towards a specific RFI source, it is unable to receive weak pulsar signals. In the interference suppression experiment, the RFI reference signal was derived from a TV antenna capturing a 4th-generation (4G) signal, featuring a 128MHz bandwidth, 16-bit complex sampling, and a center frequency of 768MHz. The bandwidth of the 4G RFI in the reference signal was 10MHz. The astronomical signals are independent of the noise in the RFI reference signals, and the amplitude and phase of the RFI vary randomly. The input and output of the adaptive filter are both complex signals. The RFI elimination experimental results are shown in Figure 2, Figure 3, and Figure 4.

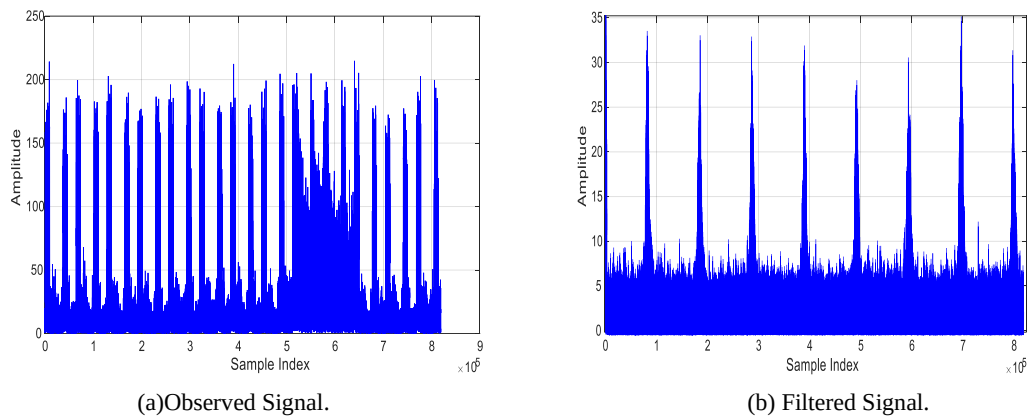


Figure 2. Time domain signals comparison before and after filtering.

Figure 2 (a) shows that astronomical signals contain strong and complex RFI, making it difficult to identify the pulsar signals inside. Figure 2 (b) shows the astronomical signal processing result after the adaptive filter with significantly improving signal-to-noise ratio (SNR), and periodic pulse signals can be clearly seen. The adaptive filter eliminates almost all RFI and recovers pulse signals.

Figure 3 shows the comparison of astronomical signals and filtered signals in the frequency domain. Figure 3 (a) shows the astronomical signals frequency domain characteristics. RFI has approximately a relatively narrow 10 MHz bandwidth, while pulsar signals have a wide 128 MHz bandwidth. Figure 3 (b) shows that the adaptive filter eliminates RFI in the 758 MHz and 768 MHz frequency bands, leaving only the pulsar signal. The pulsar signal bandwidth in the filter output is not affected.

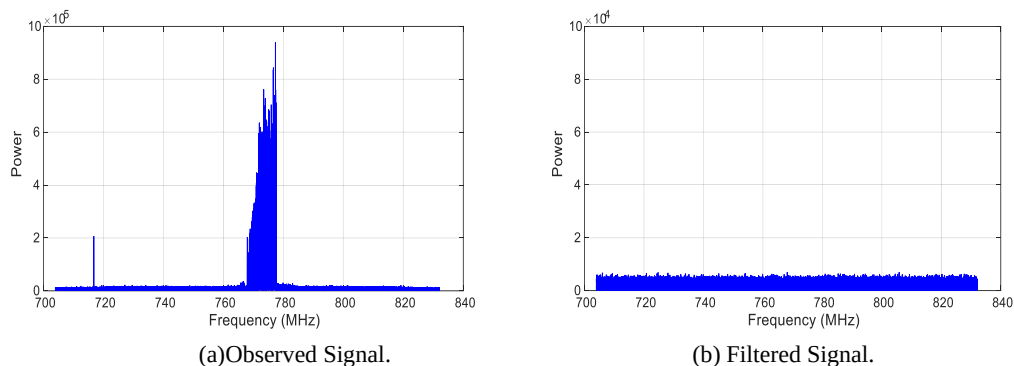
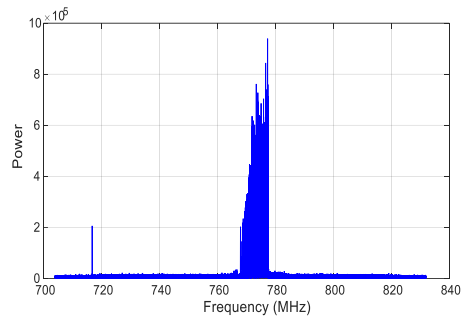


Figure 3. Frequency domain signals comparison before and after filtering.

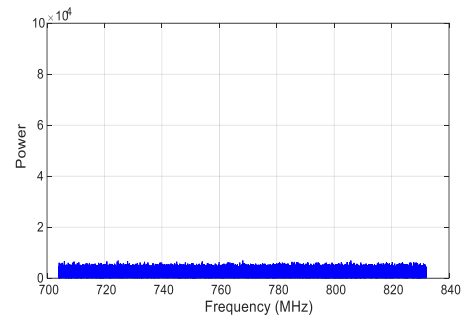
Figure 4 shows the pulse profile comparison before and after RFI elimination. Figure 4 (a) shows that the pulse profile is completely distorted by strong RFI, the pulsar signal profile cannot be obtained. Figure 4 (b) shows the clear pulsar profile after the adaptive filter processing, greatly improving the SNR.

Figures 5 and 6 show the astronomical signals 3D images before and after filtering, with the pulsar period (bins) on the x-axis, frequency channel on the y-axis, and signal power on the z-axis. The signal bandwidth is divided into 128 channels, each spanning 1 MHz.

Figure 5 demonstrates that the RFI present in the signal prior to filtering is dominant, occupying roughly ten frequency channels. The weak pulsar signal is indistinguishable, as it is entirely masked by the noise interference. Figure 6 gives the pulse signal through the adaptive filter. Around bin 230, the pulse signal emerges, spanning across all 128 frequency channels without a loss in signal bandwidth. Post-processing with the adaptive filter allows us to extract pulsar signal characteristics, including power, frequency, and period, from the astronomical signals. The adaptive filter significantly enhances the recognition rate of pulsar signals.



(a)Observed Signal Contour.



(b)Filtered Signal Contour.

Figure 4. Pulsar Signal Contour Comparison.

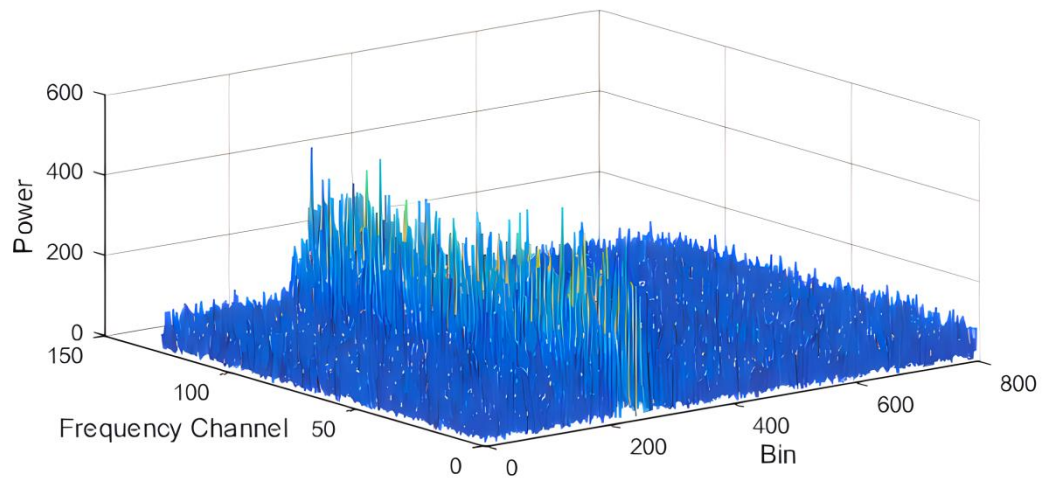


Figure 5. Observation Signal Before RFI Elimination.

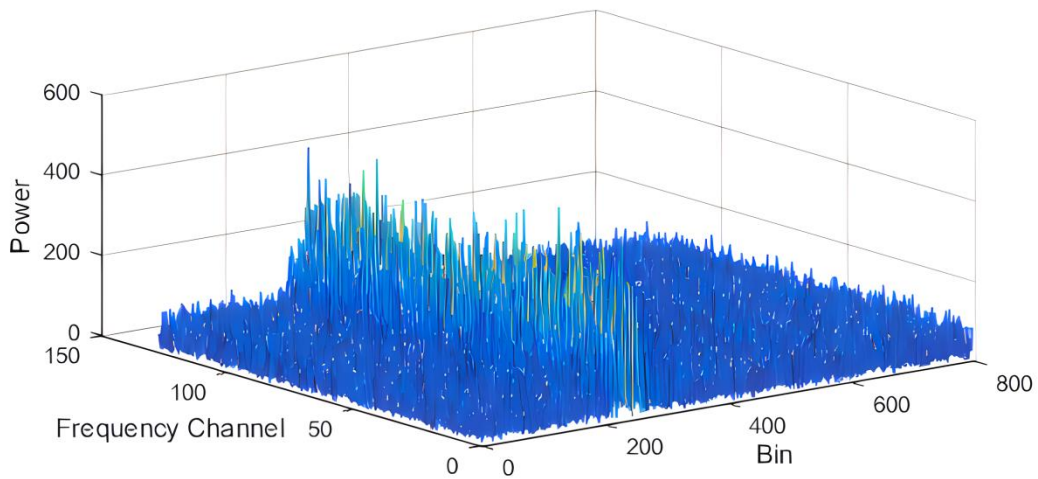


Figure 6. Observation Signal After RFI Elimination.

The filter gain is crucial in adaptive filter algorithms, determining the algorithm's RFI elimination performance. It takes time to gain from the initial state to the stable state. The filter starts to work normally after it reaches the stable state, and the gain fluctuates with the RFI change near the stable state. Figure 7 shows the adaptive filter gain change.

As shown in Figure 7, when the filter gain amplitude is 0.25, it reaches the stable state, determining the time for filters to get the stable state. If the value is small, the adaptive filter algorithm has a longer convergence time. The green line in Figure 7 shows that when $\varepsilon=0.008$, the filter takes a long time to reach a stable state, and when $\varepsilon=0.3$, the filter quickly stabilizes.

The experimental results show that the proposed adaptive filter effectively achieved complex RFI suppression, significantly improving the radio observation signals' SNR. The filter response to reach the stable state does not depend on the changes in the input signal amplitude. The interference suppression algorithm can automatically adjust the filter tap coefficients according to the RFI changes, maintaining the stable state of the filter.

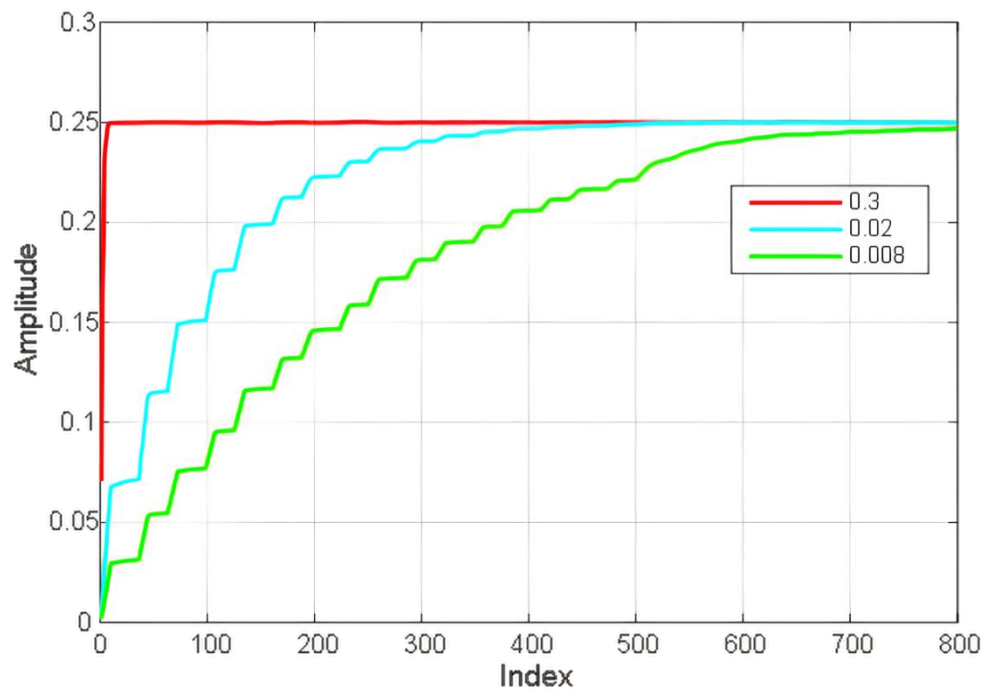


Figure 7. Filter Gain Variation Curve.

5. CONCLUSION

This article proposes an RFI suppression method based on adaptive filtering, establishes an adaptive filter model, and designs an effective RFI suppression algorithm. We received RFI signals through a reference antenna, implemented the adaptive filter using the RFI reference and the correlation with the observed signals, and eliminated complex interference mixed in the pulse signal. The RFI elimination performance of the adaptive anti-interference algorithm was verified through simulation analysis, and the SNR of the observed signal was significantly improved after adaptive filtering. The algorithm not only boosts the recognition rate and data reliability of observational signals but also holds promising potential for application in the field of radio signal observation.

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